



Can financial ratios detect fraudulent financial reporting?

Financial ratios

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Abstract Fraudulent financial reporting is a matter of grave social and economic concern. The Treadway Commission recommended that the Auditing Standards Board require the use of analytical procedures to improve the detection of fraudulent financial reporting. This is an exploratory study to determine if financial ratios of fraudulent companies differ from those of nonfraudulent companies. Fraudulent firms were identified by examining the SEC's Accounting and Auditing Enforcement Releases issued between 1982 and 1999. The fraudulent firms ($n=79$) were then matched with nonfraudulent firms on the basis of firm size, time period, and industry. Using this matched-pairs design, ratio analysis for a seven-year period (i.e. the fraud year $-/+3$ years) was conducted on 21 ratios. Overall, 16 ratios were found to be significant. Of these, only three ratios were significant for three time periods. Of the 16 statistically significant ratios, only five were significant during the period prior to the fraud year. Using discriminant analysis, misclassifications for fraud firms ranged from 58 percent to 98 percent. These results provide empirical evidence of the limited ability of financial ratios to detect and/or predict fraudulent financial reporting.

Introduction

Fraudulent financial reporting is a matter of grave social and economic concern[1]. Recent news abounds with corporate fraud scandals (e.g. Enron, WorldCom, Qwest). Such fraudulent reporting is a critical problem for external auditors, both because of the potential legal liability for failure to detect false financial statements and because of the damage to professional reputation that results from public dissatisfaction about undetected fraud. This is most recently evidenced by the demise of Arthur Andersen.

Increasing pressure to reduce fraudulent financial reporting over the past 30 years has resulted in new laws, commission reports, and standards. Congressional inquiry into the savings and loan debacle led to the formation of the Treadway Commission, whose charge was to prescribe effective recommendations to guide the Auditing Standards Board's (ASB) development of standards to help prevent and detect fraud. The Commission defined fraudulent financial reporting as intentional or reckless conduct, either by act



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or omission, that results in materially misleading financial statements (NCFR, 1987). "The Treadway Commission recommended that the ASB require the use of analytical procedures on all audits to improve the detection of fraudulent financial reporting" (Wheeler and Pany, 1996, p. 558).

In 1988, the ASB issued nine new "expectation gap" standards designed to:

- outline clearly the external auditor's role concerning fraud;
- enhance overall audit procedures for detecting and preventing fraud; and
- enhance communications between the auditor and management, the audit committee and the public (Glover and Aono, 1995).

"SAS No. 53 was designed to narrow the gap between clients' expectations regarding the auditor's responsibility to detect fraud during an audit and what that responsibility actually is" (Levy, 1989, p. 52). SAS No. 56 (AICPA, 1988) requires that analytical procedures be performed in planning the audit. In 1997, the ASB issued SAS No. 82. The standard provided expanded operational guidance on the auditor's consideration of material fraud in conducting a financial statement audit, yet had little guidance on the role of analytical procedures in fraud detection (Mancino, 1997). According to SAS No. 99, (AICPA, 2002), the new fraud standard, the results of analytical procedures should be considered in identifying the risks of material misstatement caused by fraud.

In the Fall 1997 *The Auditor's Report* of the AAA encouraged auditing practitioners and faculty to engage in research directed toward assisting auditors in preventing and detecting fraud. It suggested examining data on prior fraud litigations to find "lessons" for auditors to follow. It posited the following question – "Can analytical procedures be better used to detect fraud warning signals?" Research in this area would have been helpful to the ASB Task Force in going beyond generic guidelines to offer more specific direction to auditors (Landsittel and Bedard, 1997).

One can conclude that there is a strong need for auditing research approaches that enable the auditing practitioner to identify potential fraud. Practical evidence is needed to understand better the fraud detection capabilities of analytical procedures. The authors have chosen to accept the AAA's challenge and pursue such exploration. The focus of this study is on one commonly used analytical procedure – ratio analysis. This is a widely applied attention-direction procedure, yet little is known of the ability of ratio analysis to identify material misstatements in actual accounting data. The authors believe it is time to step back and ask a seemingly very basic question. Do financial ratios computed from the financial statements of fraudulent companies differ from those of nonfraudulent companies? Do differences exist prior to the fraud, subsequent to the fraud, both? This study is very exploratory in nature. It compares actual financial statement data of fraudulent firms to comparable nonfraudulent firms, to determine if any relationships

exist[2]. It covers multiple time periods – including both fraud and nonfraud years. The findings of the study provide the needed empirical evidence to evaluate the effectiveness of one analytical procedure – ratio analysis – in detecting fraudulent financial reporting.

A longitudinal examination of 21 financial ratios computed from annual financial statements was conducted on 79 matched pairs of firms. For each matched pair, the time period was from three years prior to the fraud year through three years post (i.e. fraud year – /+ three years). Overall, 16 ratios were found to be significant. Only three ratios were significant for three time periods and five were significant during the period prior to the fraud year. Using discriminant analysis, misclassifications for fraud firms ranged from 58 percent to 98 percent. These results provide empirical evidence of the limited ability of financial ratios to detect fraudulent financial reporting.

The remainder of this paper is organized as follows. First there is a review of the relevant research concerning analytical procedures and fraud detection. The methodology section discusses sample selection and ratios chosen for study. The results of the univariate and multivariate empirical examinations are then presented. This is followed by the study's conclusions and limitations.

Prior research

Analytical procedures (APs) have been posited to be a useful tool for identifying fraud (Thornhill, 1995). APs is the name used for a variety of techniques the auditor can use to assess the risk of material misstatements in financial records. These procedures involve the analysis of trends, ratios, and reasonableness tests derived from an entity's financial and operating data. SAS No. 56 requires that APs be performed in planning the audit with an objective of identifying the existence of unusual events, amounts, ratios and trends that might indicate matters that have financial statement and audit planning implications (AICPA, 1988). According to SAS No. 99, the current fraud standard, the auditor should consider the results of APs in identifying the risks of material misstatement due to fraud (AICPA, 2002). While the procedures are well known and widely used, there is a general lack of understanding of how they are properly applied, and how much reliance should be placed on them.

The detection of fraud in financial statements has been the subject of much empirical research. Nieschwietz *et al.* (2000) provide a comprehensive review of empirical studies related to external auditors' detection of fraudulent financial reporting. Albrecht *et al.* (2001) review the fraud detection aspects of current auditing standards and the empirical research conducted on fraud detection. The Committee of Sponsoring Organizations of the Treadway Commission sponsored a descriptive research study by Beasley *et al.* (1999) that provides a comprehensive analysis of fraudulent financial reporting occurrences investigated by the SEC subsequent to the issuance of the 1987 Treadway

Commission report. The subsequent summary discussion is limited to the use of APs in the detection of misstated financial statements. An outline of the major empirical studies is presented in Table I.

The majority of the studies involved the detection of both unintentional (i.e. errors) and intentional (i.e. irregularities/fraud) misstatements. The percent of errors identified by APs ranged from 15 percent to 50 percent (Blocher and Cooper, 1988; Calderon and Green, 1994; Hylas and Ashton, 1982; Kreutzfeldt and Wallace, 1986; Wright and Ashton, 1989). Some studies found APs not effective when done at an aggregate level (Blocher, 1992; Kinney, 1987) while others found APs not effective when applied to quarterly data (Wheeler and Pany, 1996). A wide range of APs is generally applied extensively and include both financial and operating data (Blocher, 1992; Blocher and Cooper, 1988; Daroca and Holder, 1985; Holder, 1983). There is a predominant use of simple, quantitative APs, including ratio analysis (Daroca and Holder, 1985; Kinney, 1979; Tabor and Willis, 1985; Wright and Ashton, 1989).

Only a few studies looked exclusively at detecting fraud (Blocher, 1992; Calderon and Green, 1994; Loebbecke *et al.*, 1989; Persons, 1995). Blocher (1992) found that only four out of 24 fraud cases were detected by APs. Calderon and Green (1994) found that APs were the initial signal in 15 percent of the 455 fraud cases studied. Of the AP fraud studies, only one looked at financial ratios. Persons (1995) identified 103 fraud firms which were then matched with a nonfraud firm on the basis of industry and time period. In total, ten variables, including eight ratios, were examined and used to develop two predictive logistic models. One model was for the fraud year and the other was for the preceding year. Stepwise-logistic models indicated that financial leverage, capital turnover, asset composition and firm size were significant factors influencing the likelihood of fraudulent financial reporting.

While the use of APs have been endorsed by policy makers (e.g. NCFRR, COSO, ASB, AICPA) and the subject of much research, there is little empirical evidence of their ability to detect financial statement fraud. As indicated, conclusions reached by the numerous studies have been quite disparate. Despite these findings, fraud examination texts continue to tout the use of APs and even include lists of common ratios that can be used to detect fraud (Albrecht and Albrecht, 2004; Wells, 1997). Empirical evidence for this assertion is clearly lacking. Further exploration of the ability of financial ratios to detect fraudulent firms is warranted and pursued in this study. The findings should be useful to both auditors and standard setters. Identifying key ratios useful for fraud detection may lead to a fraud detection model similar to Altman's (1968, 1983) bankruptcy prediction model and Z-score. The development of such a model is beyond the scope of this study.

	Scope of study	Objectives of study
Blocher (1992)	21 internal auditors, external auditors and controllers from 17 different firms	Examines extent APs are used by each group to detect management fraud, how effective APs are in detecting fraud, differences in usage of trend analysis, ratio analysis, and modeling among groups, experience, cognitive skills, differences in decision processes among groups
Blocher and Cooper (1988)	Five experienced auditors perform APs on 14 realistic inventory cases	Examines how AP used and if effective in detecting materially seeded errors; uses verbal protocol analysis
Calderon and Green (1994)	114 actual fraud cases published in <i>The Internal Auditor</i> , 1986-1990	Examines internal control, personnel most likely to commit fraud, personnel actions, initial signal of fraud, transaction cycle, business types
Coglitore and Berryman (1988)	Review of SEC releases and legal cases with large financial misstatements; use of operating and quarterly data	Identifies cases where APs would have revealed unusual relationships or changes in relationships and lead to detection of material misstatement
Daroca and Holder (1985)	Questionnaires sent to 1,600 members of private companies practice section of AICPA; 269 usable responses	Examines applicability and usage of different type of APs in audits and reviews
Holder (1983)	35 seniors from national, regional, local CPA firms	Examines APs selected during audit planning; case study
Hylas and Ashton (1982)	281 errors requiring adjustment on 152 audits	Examines initial event signaling error, cause of error, industry type, entity size
Kinney (1987)	48 months of actual firm data seeded with material errors; purchase not recorded or fictitious sale recorded	Examines three investigation rules: simple percentage change, statistical standardized change, pattern analysis of cross-sectional changes

(continued)

Table I.
Research on APS and
error/fraud detection

	Scope of study	Objectives of study
Kinney (1979)	Three years of data from 44 small manufacturing firms	Examines prior period adjustments as indicator of current year's errors
Kreutzfeldt and Wallace (1986)	1,506 errors for 260 engagements	Examines number and magnitude of error, types, causes, methods of detection, associations with industry, company size, SEC status and relationship to environmental factors
Loebbecke <i>et al.</i> (1989)	Survey of 121 KPMG audit partners who had experienced a material irregularity	Examines detailed information about one material irregularity selected by participant; determines presence of indicators per SAS 53 and Loebbecke-Willingham assessment model before discovery of irregularity
Persons (1995)	103 fraud firms, fraud-year period 1970-1990 matched with nonfraud firm on basis of industry and time period	Examines variables for estimating models of fraudulent financial reporting, model estimation method, and assessment of models' predictive ability
Tabor and Willis (1985)	Seven audit managers from Big Eight firm each selecting two clients and their 1978, 1982 audit work papers	Examines current role of APs in audit process, whether the process changed over the 1978-1982 period, outlook for the future
Wheeler and Pany (1996)	Five sample companies using actual data seeded with material simulated accounting errors	Examines effects of eight common errors on 15 APs (eight ratios, seven accounts); examines six models and five investigation rules
Wright and Ashton (1989)	186 engagements involving 368 proposed adjustments of a Big Eight firm	Examines types of errors, income effects, causes of errors, initial events identifying adjustments, internal control strength, ordering bias

Methodology

Identifying the population of firms involved in fraudulent financial reporting is problematic. First and foremost, fraud samples are limited to only discovered fraud. Frauds never discovered are not available for study. Frauds that are caught by the auditor and/or firm and subsequently corrected within the company are generally not revealed publicly and therefore, are similarly, not available for study. For purposes of this study, firms involved in fraudulent financial reporting were obtained from the SEC's Accounting and Auditing Enforcement Releases (AAERs) issued between 1982 and 1999. A firm reported in an AAER was included as a potential sample fraud firm if the SEC accused top management of reporting materially false and misleading financial statements. More specifically, the SEC alleged violation of Rule 10(b)-5 of the 1934 Securities Exchange Act. Rule 10(b)-5 requires the intent to deceive, manipulate or defraud. On finding sufficient evidence of fraud, the court entered final judgment of permanent injunction.

Auditing standards (AICPA, 1997, 2002) and prior research (Beasley *et al.*, 1999; Loebbecke *et al.*, 1989; Sorenson *et al.*, 1983) have found that accounts receivable and inventory are important variables in assessing fraud risk. Beasley *et al.* (1999) found these were the two most common misstated asset accounts. To ensure investigation of these variables, banking and insurance firms were excluded from the sample.

For purposes of this study, the "fraud year" was defined as the first year for which the financial statements included fraudulent data. In most instances, the actual discovery of the fraud occurred several years subsequent to the fraud year. According to the Wells Report (NCFRR, 1987), financial statement fraud occurred for an average of 36 months before being detected. The financial statement data of a fraud firm were collected from SEC 10K and/or annual report files. To reduce hindsight bias, the original financial statements, rather than restated financials, were used. See Table II for a reconciliation of AAERs and the identification of fraud firms.

Using COMPUSTAT, each fraud firm was matched with a nonfraud firm based on the three following requirements:

- (1) *Firm size.* A nonfraud firm was considered similar if total assets were within $-/+$ 30 percent of total assets for the fraud firm in the year

Accounting and Auditing Enforcement Releases (AAERs)	
#1 – 1,148 for the period 4/82 to 8/99	1,148
Less:	
AAERs not referencing violation of Rule 10(b)-5	(619)
AAERs affecting banks/insurance firms, CPA firms, 10Qs, registration statements, or fraud year(s) not identified	(304)
AAERs expanding other AAERs (e.g. duplicate AAERs for same firm)	(146)
Total number of fraud firms included in study	79

Table II.
Identification of fraud firms

preceding the fraud year (i.e. year-1); if no match was found, $-/+ 30$ percent of total sales was used.

- (2) *Time period.* Firms identified in (1) above were reviewed to identify those nonfraud firms for which 10K and/or annual reports were available for the same time period as the fraud firm.
- (3) *Industry.* Firms identified in (2) above were reviewed to identify a nonfraud firm within the same four-digit SIC as the fraud firm; the nonfraud firm chosen was the one with the closest total assets or total sales to the fraud firm; if no match was found, three-digit codes were used.

There were several limitations of the above sample selection process. First and foremost was the inability to identify all companies engaged in fraudulent financial reporting. The identified fraud firms were limited to public firms with discovered fraud that was publicly revealed and pursued by SEC enforcement actions. Another limitation was the possible misclassification of a nonfraud firm. Financial statement fraud might have occurred but had not been detected, publicly revealed and/or subject to SEC investigation.

To determine if differences exist prior to the fraud and/or subsequent to the fraud, a longitudinal examination was necessary. For each fraud and nonfraud firm, financial statement data were collected for the fraud year, three years prior through three years post (i.e. fraud year $-/+$ three years). The financial statements used included the original data rather than any resulting subsequent restatements. Given that a majority of fraud firms cease operations (e.g. declare bankruptcy), using a longer time period resulted in an inadequate sample size. Even using a total time span of seven years, there were instances where complete data were not available (e.g. firm not public entire period, bankruptcy proceedings). Accordingly, the sample size varied for both fraud firms and nonfraud firms for each time period. The fraud years ranged from 1978 to 1996, giving a total time period from 1975 to 1999.

The variables used in this study were ratios computed from the sample firms' income statements and balance sheets. Ideally, the selection of financial ratios used for analysis should be based on theory and coupled with demonstrated empirical evidence of their usefulness. An acceptable theoretical foundation for the selection of ratios for decision making does not currently exist. Prior studies have produced scattered heterogeneous empirical evidence regarding ratio usefulness (Deakin, 1976; Persons, 1995; Pinches *et al.*, 1973, 1975; Wheeler and Pany, 1996) To date, a complete set of useful ratios has not been identified. Given the time frame of this study, (i.e. mid-1970s thru late 1990s) and that the statement of cash flows was not a required financial statement until 1987, ratios based on cash flow data were not included. Given these considerations and the exploratory nature of this study, a fairly comprehensive set of financial ratios was chosen for inclusion. Since published

financial statements were the data source, ratios using internal operational-type measures were also excluded. See Table III for a complete listing of ratios.

The methodology of this study was similar to the one used by Persons (1995). That study compared fraud versus nonfraud firms for a two-year period – the fraud year and the preceding year. Ten variables were used including eight ratios. Size was used as a variable rather than a sample-matching criterion. This study was much more comprehensive, looking at many more ratios over a more extended period of time. This study also performed univariate analysis on each ratio for each time period. Whereas the Persons (1995) study involved a logistic prediction model, this study performed discriminant analysis.

Results

Both univariate and multivariate analyses were conducted. Given the matched-pairs design, paired-sample *t* tests were conducted for each variable (i.e. ratio) to determine if the mean of the fraud sample was significantly different than the mean of the nonfraud sample. Results of the *t* tests are reported in Table IV.

Of the 21 ratios analyzed for the seven-year time period, 16 were found to be statistically significant at an α level of 0.10 or less. Of these, nine were significant for only one time period and four were significant for two time periods. Only three ratios were significant for three time periods (i.e. FA/TA,

^a V1	AR/TA	Accounts receivable/total assets
V2	INVTO	Cost of goods sold/inventory
V3	COGS/SAL	Cost of goods sold/sales
V4	CURRAT	Current assets/current liabilities
^a V5	CA/TA	Current assets/total assets
V6	DE	Total liabilities/total equity
V7	FA/TA	Fixed assets/total assets
V8	GP%	Gross profit/sales
V9	IE/TL	Interest expense/total liabilities
V10	INC/CA	Inventory/current assets
V11	INV/SAL	Inventory/sales
^a V12	INV/TA	Inventory/total assets
V13	NI/SAL	Net income/sales
^a V14	NI/TA	Net Income/total assets
V15	OPX/SAL	Operating expenses/sales
V16	OPI/SAL	Operating income/sales
^a V17	RE/TA	Retained earnings/total assets
V18	SAL/AR	Sales/accounts receivable
^a V19	SAL/TA	Sales/total assets
^a V20	TL/TA	Total liabilities/total assets
^a V21	WC/TA	Working capital/total assets

Note: ^aRatio used in Persons (1995)

Table III.
Ratios

Table IV.
Paired *t*-tests – fraud vs
nonfraud *p* values

		Year – 3	Year – 2	Year – 1	Fraud year	Year +1	Year +2	Year +3
V1	AR/TA	0.2904	0.0881*	0.9739	0.7249	0.8670	0.2056	0.0344**
V2	INVTO	0.2342	0.3113	0.5810	0.8021	0.3496	0.2701	0.3653
V3	COGS/SAL	0.0983*	0.2368	0.2113	0.4943	0.1439	0.0213**	0.6629
V4	CURRAT	0.4961	0.4735	0.6169	0.6884	0.1231	0.0212**	0.1588
V5	CA/TA	0.6317	0.1337	0.4217	0.3388	0.0695*	0.1609	0.5952
V6	DE	0.2943	0.3591	0.6459	0.2918	0.2175	0.3774	0.5039
V7	FA/TA	0.1259	0.0412**	0.0031***	0.0001***	0.6325	0.1269	0.1447
V8	GP%	0.8298	0.7448	0.9749	0.4817	0.8879	0.2466	0.7944
V9	IE/TL	0.0652***	0.3972	0.3792	0.1686	0.3048	0.3224	0.3290
V10	INV/CA	0.2005	0.6256	0.1576	0.0679**	0.6942	0.5778	0.3324
V11	INV/SAL	0.2684	0.3264	0.3196	0.0830**	0.3657	0.6976	0.2902
V12	INV/TA	0.4030	0.9129	0.3435	0.4003	0.4840	0.6094	0.6761
V13	NI/SAL	0.3526	0.9879	0.4329	0.3948	0.2497	0.1071	0.3554
V14	NI/TA	0.2743	0.3166	0.2175	0.2950	0.2503	0.0004***	0.0107**
V15	OPX/SAL	0.3571	0.8800	0.5829	0.3091	0.3328	0.0080***	0.1066
V16	OPI/SAL	0.3604	0.8115	0.2749	0.6963	0.7864	0.0041***	0.2819
V17	RE/TA	0.5029	0.3157	0.5077	0.7753	0.1956	0.0086***	0.0005***
V18	SAL/AR	0.1471	0.2444	0.0568*	0.1050	0.7957	0.5723	0.2198
V19	SAL/TA	0.2088	0.7166	0.1696	0.0463**	0.1821	0.5684	0.3363
V20	TL/TA	0.3080	0.2837	0.7156	0.4319	0.0845*	0.0134**	0.0003***
V21	WC/TA	0.1994	0.5010	0.7074	0.6771	0.0254**	0.0022***	0.0061***

Note: * Significant at $\alpha = 0.10$; ** Significant at $\alpha = 0.05$; *** Significant at $\alpha = 0.01$

TL/TA, WC/TA). Of the 16 statistically significant ratios, only five were significant during the period prior to the fraud year (i.e. AR/TA, COGS/SAL, FA/TA, IE/TL, SAL/AR), with only one ratio significant for both year two and year one (i.e. FA/TL). There were two ratios consistently significant during the period subsequent to the fraud year (i.e. TL/TA, WC/TA) and two ratios were significant for both year+2 and year+3 (i.e. NI/TA, RE/TA). These results provide empirical evidence of the limited ability of financial ratios to detect and/or predict the occurrence of fraud.

Given the large number of independent variables (i.e. 21 ratios), multivariate analysis was warranted. Discriminant analysis is an appropriate statistical technique for testing the hypothesis that the group means of a set of independent variables for two groups (e.g. fraud vs nonfraud) are equal. For each of the seven years tested, the chi-square value to test the homogeneity of within covariance matrices was found to be significant. Accordingly, the within covariance matrices were used in calculating a quadratic discriminant function. Results are reported in Table V.

The resubstitution method used the same data both to define and to evaluate the classification criterion. As shown in Panel A, the ratios were quite effective in classifying nonfraud firms, having a minimum hit ratio of 98 percent and misclassifying a firm as fraudulent less than 2 percent of the time. Meanwhile,

					Financial ratios
		Panel A: resubstitution method		Panel B: cross-validation method	
		F (%)	N (%)	F (%)	N (%)
<i>Year - 3</i>					
F		36.59	63.41	7.32	92.68
N		0.00	100.00	15.79	84.21
<i>Year - 2</i>					
F		23.64	76.36	1.82	98.18
N		1.54	98.46	13.85	86.15
<i>Year - 1</i>					
F		25.37	74.63	19.40	80.60
N		1.43	98.57	14.29	85.71
<i>Fraud year</i>					
F		27.54	72.46	21.74	78.26
N		1.41	98.59	14.08	85.92
<i>Year +1</i>					
F		34.55	65.45	21.82	78.18
N		0.00	100.00	11.27	88.73
<i>Year +2</i>					
F		51.16	48.84	41.86	58.14
N		0.00	100.00	10.00	90.00
<i>Year +3</i>					
F		58.82	41.18	17.65	82.35
N		1.43	98.57	12.86	87.14

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Table V.
Discriminant analysis

the hit ratio for fraud firms was between a low of 24 percent and a high of 59 percent. Misclassifications ranged from 41 percent to 76 percent.

To reduce the optimistic bias resulting from the resubstitution method, two options were available. The first option was to split the sample into two sets, using one part to derive the discriminant function and the other set to test the classification criterion. Alternately, the cross-validation procedure (Lachenbruch and Mickey, 1968) can be used whereby one observation is left out when determining the discriminant function and then used for classification purposes. This procedure is then repeated for each of the observations. The final result is a nearly unbiased estimate.

Given the limited number of fraud firms available, the split sample option was not feasible. Accordingly, the cross-validation method was used and the results are presented in Panel B. As expected, the hit ratios declined for both the fraud firms and the nonfraud firms. The tested financial ratios were still effective at identifying the nonfraud firms, having a misclassification range of between 10 percent and 16 percent. Meanwhile, the fraud firms were misclassified as nonfraudulent between 58 percent and 98 percent of the time. These results provide additional empirical evidence of the limited ability of financial ratios to detect and/or predict fraudulent financial reporting.

Conclusion

APs are used to assess the risk of material misstatements in financial records. The Treadway Commission even recommended that the ASB require the use of APs on all audits as a means to detect fraudulent financial reporting (Wheeler and Pany, 1996). According to SAS No. 99, the auditor should consider the results of APs in identifying fraud risk (AICPA, 2002). There is a widely held contention one common AP, ratio analysis, is a useful tool for identifying fraud. The purpose of this study was to explore the fraud detection capabilities of ratio analysis. It compared a multitude of financial ratios for matched fraudulent and nonfraudulent firms to see if differences existed. It examined an extended time period both pre- and post-fraud years. Statistically, there was not much difference in the ratios of fraud versus nonfraud firms. Those ratios found significant were not consistent across the time periods. A discriminant prediction model misclassified fraud firms from 58 percent to 98 percent of the time.

The results of this study provide empirical evidence of the limited ability of ratio analysis to detect fraudulent financial reporting. Such findings should be useful to both standard setters and auditors in their prescription and application of ratio analysis for the detection of financial statement fraud.

This study only partially addressed the question posed by the AAA – “Can analytical procedures be better used to detect fraud warning signals?” (Landsittel and Bedard, 1997, p. 4). Additional research regarding the appropriate use of APs to detect fraud is urgently needed for the prescription of more effective detailed auditing standards.

There were several limitations to this study. First, the identified fraud firms were limited to public firms with discovered fraud that were subject to SEC enforcement actions. Second, a nonfraud firm might have been misclassified. Financial statement fraud might have occurred but had not been detected or subject to SEC investigation. Owing to the low incidence of financial statement fraud and the need for an adequate sample size, the sample extends over a long period. Such maturation often results in changing conditions within the sample period which can also impact the model data and the prediction period. Owing to sample size limitations, the use of a “hold out” sample to validate the discriminant model was not feasible. Finally, there does not exist an acceptable theoretical foundation for the selection of financial ratios for decision making. The ratios selected for inclusion in this study were based on scattered heterogeneous empirical evidence and logical inferences of accounts most likely involved in fraudulent financial reporting. Different results might ensue if different ratios were selected.

Notes

1. While the term “fraud” in prior auditing standards referred to irregularities which incorporated fraudulent financial reporting as well as employee theft, embezzlement or defalcation, we limit our focus to management fraud or fraudulent financial reporting, which

relates primarily to management's intentional misrepresentation in or omission from financial statements. A similar definition of fraud is used by Nieschwietz et al. (2000), who provide a review of the empirical research on external auditors' detection of financial statement fraud.

2. Data available from public sources; for more information, please contact the first author.

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